

Justify all answers. Submit as a single PDF.

- (10 points) In class we saw how linear probing can be used to resolve collisions. Another way to resolve collisions in hashing is to use *double hashing*. We use an initial hashing function $h(k) = k \bmod p$, where p is prime. We also use a second hashing function $g(k) = (k + 1) \bmod (p - 2)$. When a collision occurs, we use a probing sequence

$$h(k, i) = (h(k) + i \cdot g(k)) \bmod p.$$

Use the double hashing procedure we have described with $p = 4969$ to assign memory locations to files for employees with social security numbers $k_1 = 132489971$, $k_2 = 509496993$, $k_3 = 546332190$, $k_4 = 034367980$, $k_5 = 047900151$, $k_6 = 3229938157$, $k_7 = 212228844$, $k_8 = 325510778$, $k_9 = 353354519$, $k_{10} = 053708912$.

- (10 points) **The power generator** is a method for generating pseudorandom numbers. To use the power generator, parameters p and d are specified, where p is a prime, d is a positive integer such that $p \nmid d$, and a seed x_0 is specified. The pseudorandom numbers $x_1, x_2, \dots$ are generated using the recursive definition

$$x_{n+1} \equiv x_n^d \pmod{p}.$$

Find the sequence of pseudorandom numbers generated by the power generator with $p = 7$, $d = 3$, and seed $x_0 = 2$.

- Periodicals are identified using an **International Standard Serial Number (ISSN)**. An ISSN consists of two blocks of four digits. The last digit in the second block is a check digit. This check digit is determined by the congruence

$$d_8 \equiv 3d_1 + 4d_2 + 5d_3 + 6d_4 + 7d_5 + 8d_6 + 9d_7 \pmod{11}.$$

When $d_8 \equiv 10 \pmod{11}$, we use the letter X to represent d_8 in the code.

- (5 points) Determine the ISSN check digit for (may be the letter X):
(i) 1570-868 (ii) 1553-734 (iii) 1089-708 (iv) 1383-811
 - (5 points) Are each of these eight-digit codes possible ISSNs?
(i) 1059-1027 (ii) 0002-9890 (iii) 1530-8669 (iv) 1007-120X
 - (5 points) Determine which single errors the check digit of an ISSN detects.
 - (5 points) Determine which transposition errors the check digit of an ISSN detects.
- Encrypt the message WATCH YOUR STEP by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters.
 - (5 points) $f(p) = (p + 14) \bmod 26$
 - (5 points) $f(p) = (14p + 21) \bmod 26$
 - (5 points) $f(p) = (-7p + 1) \bmod 26$

5. Decrypt these messages encrypted using the shift cipher $f(p) = (p + 10) \bmod 26$.
- (a) (5 points) CEBBOXNOB XYG
 - (b) (5 points) LO WI PBSOXN
 - (c) (5 points) DSWO PYB PEX
6. (10 points) Suppose that the ciphertext ERC WYJJMG MIRXPC EHZERGIH XIGLRSPSKC MW MRHMWXMRKYM WLEFPI JVSQ QEKMG was produced by encrypting a plaintext message using a shift cipher. What is the original plaintext?
7. (10 points) Caesar wants to arrange a secret meeting with Marc Antony, either at the Tiber (the river) or at the Coliseum (the arena). He sends the ciphertext EVIRE. However, Antony does not know the key, so he tries all possibilities. Where will he meet Caesar? (*Hint: This is a trick question.*)
8. (10 points) Suppose you encrypt using an affine cipher, then encrypt the encryption using another affine cipher (both are working mod 26). Is there any advantage to doing this, rather than using a single affine cipher? Why or why not?
9. Using the Vigenère cipher, answer the following:
- (a) (5 points) Use key BLUE to encrypt the message SNOWFALL.
 - (b) (5 points) The ciphertext OIKYVWHBX was produced by encrypting a plaintext using key HOT. What is the plaintext?
10. Suppose there is a language that has only the letters a and b . The frequency of the letter a is 0.1 and the frequency of b is 0.9. A message is encrypted using a Vigenère cipher (working mod 2 instead of mod 26). The ciphertext is BABABAAABA.
- (a) (5 points) Show that the key length is probably 2.
 - (b) (10 points) Using the information on the frequencies of the letters, determine the key and decrypt the message.
11. The operator of a Vigenère encryption machine is bored and encrypts a plaintext consisting of the same letter of the alphabet repeated several hundred times. The key is a six-letter English word. Eve knows that the key is a word but does not yet know its length.
- (a) (5 points) What property of the ciphertext will make Eve suspect that the plaintext is one repeated letter and will allow her to guess that the key length is six?
 - (b) (5 points) Once Eve recognizes that the plaintext is one repeated letter, how can she determine the key? (*Hint: Assume no English word of length six is a shift of another English word.*)
 - (c) (5 points) Suppose Eve doesn't notice the property needed in part (a), and therefore uses the method of displacing then counting matches for finding the length of the key. What will the number of matches be for the various displacements? In other words, why will the length of the key become very obvious by this method?